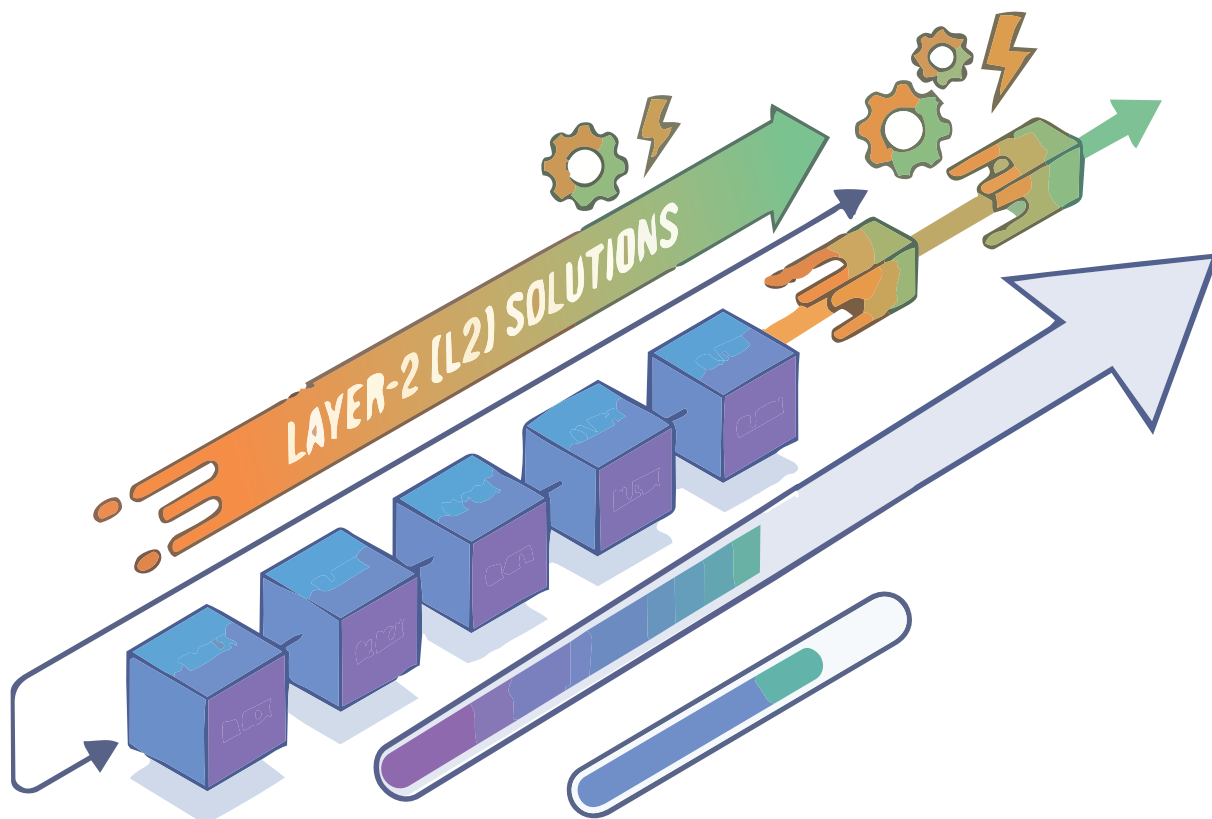


How Layer-2 Solutions are Transforming the Blockchain

Layer-2 solutions are reshaping the speed and cost of blockchain technology for the better.



Imagine critical business deals being made on a one-lane highway at a time when traffic is at its peak. Vehicles will crawl, fuel costs will mount, tempers will flare, and profit margins will fall as precious moments are spent on the road instead of at work. Public blockchains such as Ethereum have encountered this problem since their inception. Basic blockchain activities like value transfer and dApp (decentralised application) usage cost more than ten dollars while users experience wait times that last between thirty minutes to several hours during periods of network congestion.

The year 2026 marks the end of this bottleneck. Layer-2 (L2) technologies can handle thousands of transactions using their off-chain processing system while maintaining security through a strong Layer-1 security framework. The data shows that L2 networks currently process approximately 2 million

transactions per day, which is double the transaction volume of the Ethereum Mainnet. The total value locked (TVL) across major Layer 2 networks exceeds the \$40 billion mark, while some networks reach \$50 billion. Major players like Arbitrum and Base have attracted billions in TVL while they support both decentralised finance (DeFi) applications and consumer software applications at costs of less than one cent.

Layer-2 technology enables blockchain technology to move from its experimental stage into its practical deployment phase, creating operational systems that support businesses.

Why blockchain felt like a crowded highway at rush hour

Layer-1 networks established security and decentralisation as their main goals but this design choice restricted their ability